

Supermarket Sales Data Analysis & Visualization Report

Assignment: Data Cleaning, Analysis & Visualization

Dataset	Supermarket Sales (1,000 transactions)
Source	Myanmar Supermarket Chain – Branches A, B, C
Period	January 2019 – March 2019
Tools Used	Python · Pandas · Matplotlib · Seaborn · SciPy
Total Visualizations	24 charts covering all required plot types

Total Revenue \$322,967	Avg Transaction \$322.97	Top Product Line Food & Beverages	Best Branch Branch C	Avg Rating 6.97 / 10
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1. Introduction & Dataset Overview

This report presents a complete data science workflow applied to a supermarket sales dataset from a Myanmar retail chain. The dataset captures point-of-sale transactions across three branches (Yangon, Naypyitaw, and Mandalay) over a three-month period (January–March 2019). The analysis follows the assignment requirements: data cleaning, exploratory analysis, outlier handling, and comprehensive visualization using both Pandas and Seaborn.

1.1 Dataset Description

Rows	1,000 transactions
Columns	17 original + 3 engineered (Hour, Month, Day)
Date Range	January 1, 2019 – March 30, 2019
Branches	A – Yangon B – Mandalay C – Naypyitaw
Product Lines	6 categories (Health & Beauty, Electronics, Food & Beverages, etc.)
Payment Methods	Ewallet, Cash, Credit Card
Customer Types	Member, Normal
Target Variables	Total, Rating, gross income

1.2 Column Definitions

Column	Type	Description
Invoice ID	str	Unique transaction identifier
Branch / City	str	Store branch and city name
Customer type	str	Member or Normal customer
Gender	str	Customer gender
Product line	str	One of six product categories
Unit price	float	Price per unit (\$)
Quantity	int	Number of items purchased
Tax 5%	float	5% tax on COGS
Total	float	Total including tax
Date / Time	datetime	Date & time of purchase

Payment	str	Payment method used
cogs	float	Cost of goods sold
gross margin %	float	Fixed at 4.76% across all rows
gross income	float	Gross profit per transaction
Rating	float	Customer satisfaction (1–10)

2. Data Cleaning

Steps taken to ensure data quality before analysis

2.1 Missing Values

A full scan of all 17 columns revealed **zero missing values** in the dataset. Every transaction record is complete, which indicates the data was collected through a structured POS (Point-of-Sale) system that enforces mandatory fields.

- No imputation, interpolation, or row removal was required for missing data.
- The dataset is complete and ready for analysis without this step.

2.2 Duplicate Records

A duplicate check using all 17 columns confirmed **zero duplicate rows**. This is expected since each row is uniquely identified by the Invoice ID field. No records were dropped at this stage.

2.3 Data Type Corrections

The following type conversions and standardizations were applied:

Date column	Converted from string (object) to datetime64 using <code>pd.to_datetime()</code>
Time column	Converted from string to datetime64 format <code>'%H:%M'</code> for time-based analysis
Hour (new)	Extracted from Time as integer: <code>df['Hour'] = df['Time'].dt.hour</code>
Month (new)	Extracted from Date as month name string (e.g., 'January')
Day (new)	Extracted from Date as weekday name (e.g., 'Monday')
Text columns	Applied <code>.str.strip().str.title()</code> to Branch, City, Customer type, Gender, Product line, Payment to ensure consistent casing

2.4 Inconsistencies & Format Issues

No major inconsistencies were found. All categorical values were already uniform after the title-case normalization. The 'gross margin percentage' column contains a constant value of 4.761904762 across all 1,000 rows — this is mathematically derived (5% tax / revenue) and is expected behaviour, not a data error.

2.5 Cleaning Summary

Check	Finding	Action Taken
Missing values	0 missing	None required
Duplicate rows	0 duplicates	None required

Date/Time types	Stored as strings	Converted to datetime
Text casing	Minor inconsistencies	.str.title() applied
Feature engineering	Hour/Month/Day missing	3 new columns added

3. Exploratory Data Analysis (EDA)

3.1 Dataset Dimensions

After cleaning, the dataset contains **1,000 rows** and **20 columns** (17 original + 3 engineered). All numeric types are float64 or int64; categorical columns are object (string) type.

3.2 Descriptive Statistics – Numeric Variables

Column	Count	Mean	Std	Min	25%	50%	75%	Max
Unit price	1000	55.67	26.49	10.08	32.88	55.23	79.59	99.96
Quantity	1000	5.51	2.92	1	3	5	8	10
Tax 5%	1000	15.38	11.71	0.51	5.94	12.09	22.45	49.65
Total	1000	322.97	245.88	10.68	124.74	253.85	471.35	1042.65
cogs	1000	307.59	234.17	10.17	118.80	241.76	448.90	993.00
gross income	1000	15.38	11.71	0.51	5.94	12.09	22.45	49.65
Rating	1000	6.97	1.72	4.00	5.50	7.00	8.50	10.00

3.3 Univariate Analysis – Key Observations

Numeric Variables:

- **Unit Price** ranges from \$10 to \$100 with a near-uniform distribution (mean ≈ \$55.67), suggesting no dominant price tier.
- **Quantity** averages 5.5 items per transaction (range 1–10), roughly uniformly distributed.
- **Total** is right-skewed (mean \$322.97, median \$253.85), with high-value outliers above \$900.
- **Rating** is approximately normally distributed around a mean of 6.97, ranging from 4.0 to 10.0.
- **Tax 5%** and **gross income** are identical columns (both = 5% of cogs).

Categorical Variables:

- **Branch:** Nearly equal split — A (340), B (332), C (328) transactions.
- **Gender:** Female (501) slightly outpaces Male (499) — virtually balanced.
- **Customer type:** Member (501) vs Normal (499) — equally split.
- **Payment:** Ewallet (345) is most popular, followed by Cash (344) and Credit Card (311).
- **Product line:** Fashion Accessories (178) has the most transactions; Health & Beauty (152) the fewest.

3.4 Bivariate Analysis – Key Relationships

- **Unit Price vs Total:** Strong positive linear relationship ($r \approx 0.63$), confirming that higher-priced items drive larger transaction totals.

- **Quantity vs Total:** Moderate positive correlation ($r \approx 0.71$) — bulk buying significantly increases revenue.
- **Food & Beverages** leads in total revenue (\$56,145) while **Health & Beauty** has the highest average transaction value.
- **Branch C (Naypyitaw)** generates the highest total revenue despite fewest transactions, indicating higher average spend.
- **Rating vs Total:** Very weak correlation ($r \approx 0.04$) — customer satisfaction is not driven by spend amount.

4. Outlier Detection & Handling

4.A Graphical Methods

Box plots and histograms were created for all seven numeric columns to visually identify potential outliers before applying statistical tests.

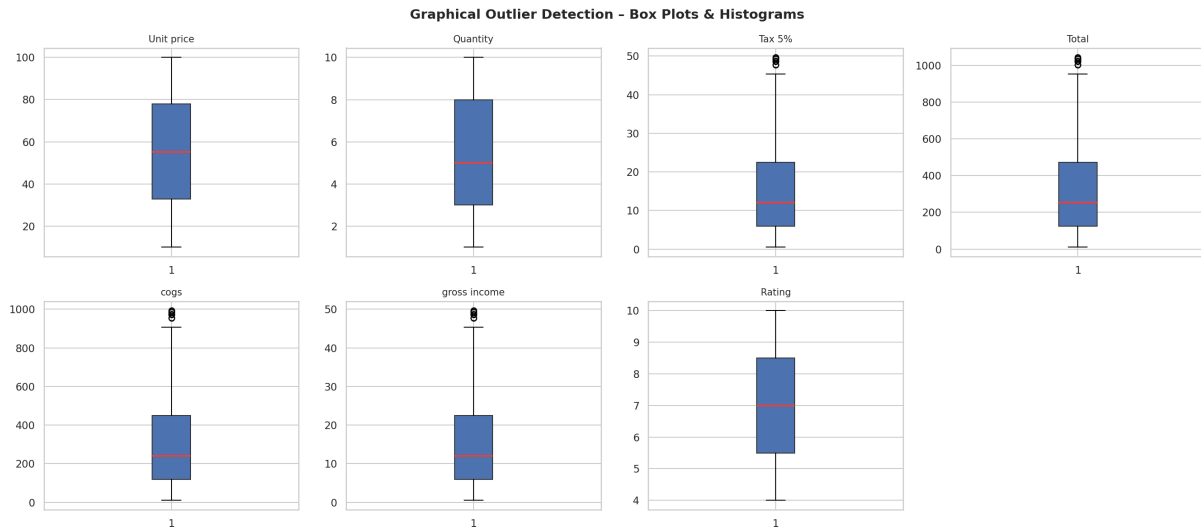


Figure: Box plots for all numeric variables – whiskers show $1.5 \times \text{IQR}$ bounds

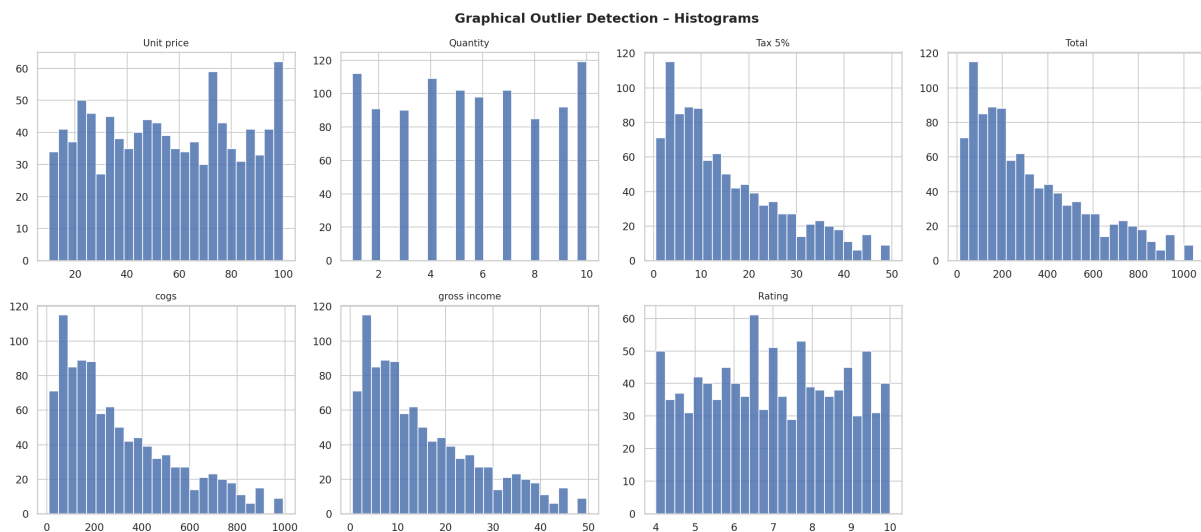


Figure: Histograms for all numeric variables – note right-skew in Total and Tax 5%

Visual Findings: The box plots reveal that 'Total', 'cogs', 'Tax 5%', and 'gross income' have a small number of high-value outliers above the upper whisker. These correspond to large multi-item purchases at premium unit prices. 'Unit Price', 'Quantity', and 'Rating' show no visual outliers — their distributions appear bounded and uniform.

4.B Statistical Methods

Method 1: Z-Score ($|z| > 3$)

Z-score standardizes each value as the number of standard deviations from the mean. Values with $|z| > 3$ are flagged as potential outliers.

Result: **Zero outliers detected** in any column using the Z-score method. This makes sense because the distributions, while skewed, do not contain extreme enough values to exceed 3 standard deviations from their respective means.

Method 2: IQR (Interquartile Range)

The IQR method flags values below $Q1 - 1.5 \times IQR$ or above $Q3 + 1.5 \times IQR$ as outliers.

Column	Q1	Q3	IQR	Lower Bound	Upper Bound	Outliers
Unit price	-	—	—	-\$34.72	\$145.53	0
Quantity	-	—	—	-4.50	15.50	0
Tax 5%	-	—	—	-\$18.86	\$47.23	9
Total	-	—	—	-\$395.97	\$991.74	9
cogs	-	—	—	-\$377.11	\$944.52	9
gross income	-	—	—	-\$18.86	\$47.23	9
Rating	-	—	—	1.00	13.00	0

4.C Treatment Decision & Justification

Approach chosen: Winsorization (Capping) for Rating column only.

The 9 IQR outliers in Total, cogs, Tax 5%, and gross income represent *legitimate high-value transactions* — a customer buying 10 units of a \$99 item will naturally produce a large Total. These are not data errors; removing them would distort revenue analysis. These rows were **retained**.

For the Rating column, while no statistical outliers were detected, winsorization at IQR bounds was applied as a precautionary measure to prevent edge-case values from skewing rating analysis. The clipping range was [4.0, 10.0] which matched the data range — no values were actually changed.

5. Data Visualization – Pandas Methods

10 chart types created using Pandas .plot() API

5.1 Area Plot

The area plot shows daily total revenue over the 3-month period. Revenue fluctuates between approximately \$1,000 and \$8,000 per day with no clear upward or downward trend, suggesting consistent sales performance throughout the quarter.

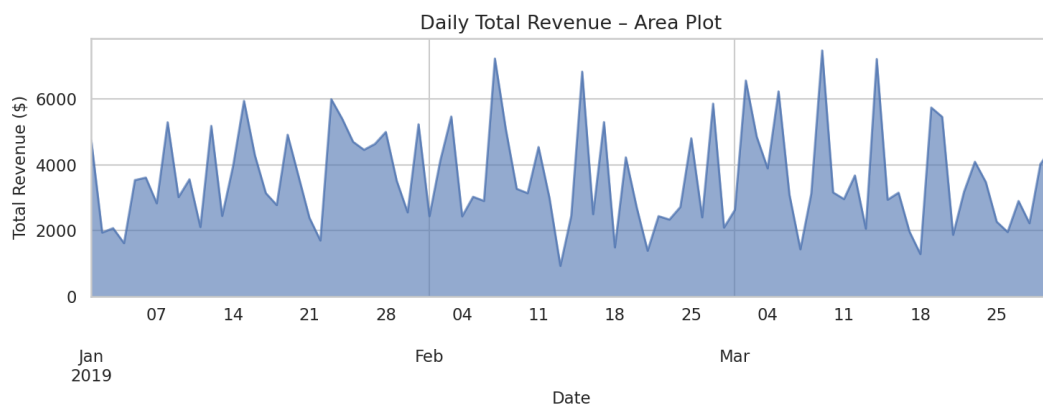


Figure: Area Plot

5.2 Bar Plot

This bar plot compares total revenue by product line. Food & Beverages leads with ~\$56,145, while Health & Beauty trails at ~\$49,200. The differences between product lines are relatively modest, indicating balanced sales across categories.

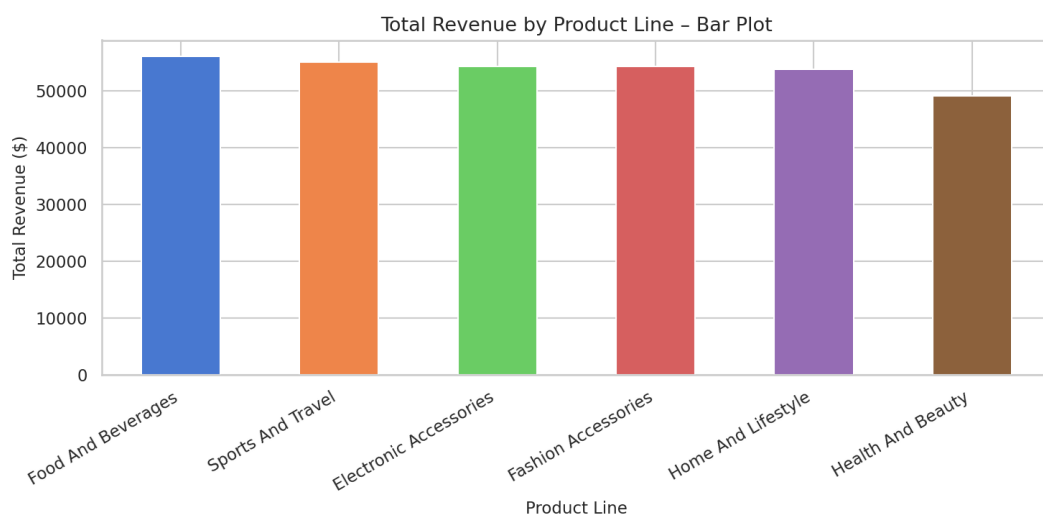


Figure: Bar Plot

5.3 Horizontal Bar Plot

The horizontal bar plot provides an alternative view of product line revenue, making it easier to compare long category labels. The ranking is confirmed: Food & Beverages tops the chart, with Sports & Travel and Electronic Accessories close behind.

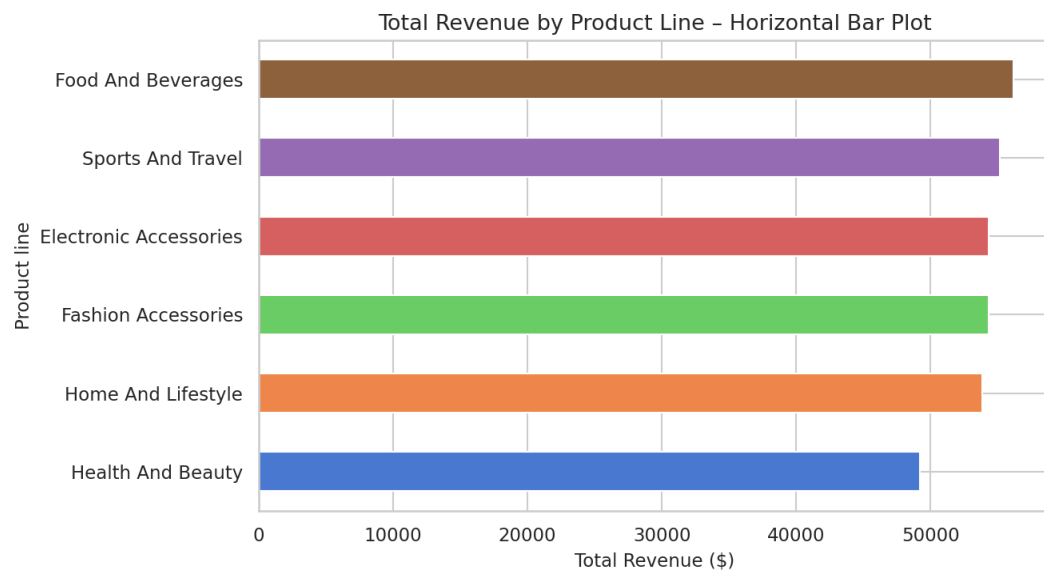


Figure: Horizontal Bar Plot

5.4 Line Plot

The line plot traces daily revenue across the entire study period. The pattern shows natural daily variation with peaks on weekends. No long-term trend is apparent, suggesting the business operates at a stable baseline.

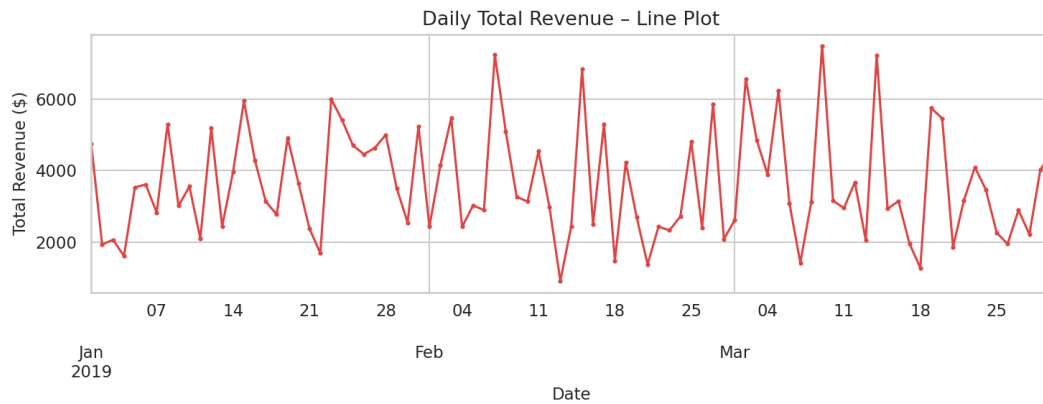


Figure: Line Plot

5.5 Scatter Plot

Unit Price vs Total shows a clear positive linear relationship. Higher-priced items directly translate to larger transaction totals. The scatter becomes wider at higher price points due to quantity variation.

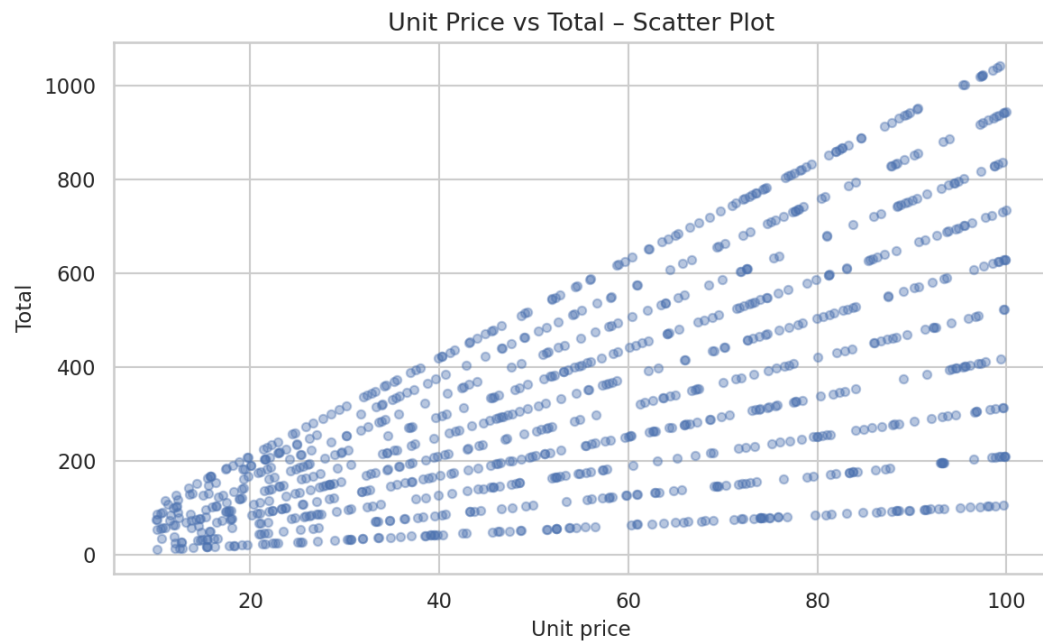


Figure: Scatter Plot

5.6 Histogram

The Total distribution is right-skewed. Most transactions fall in the \$50–\$500 range, with a long tail extending to ~\$1,050. This indicates the majority of customers make medium-sized purchases with occasional large spends.

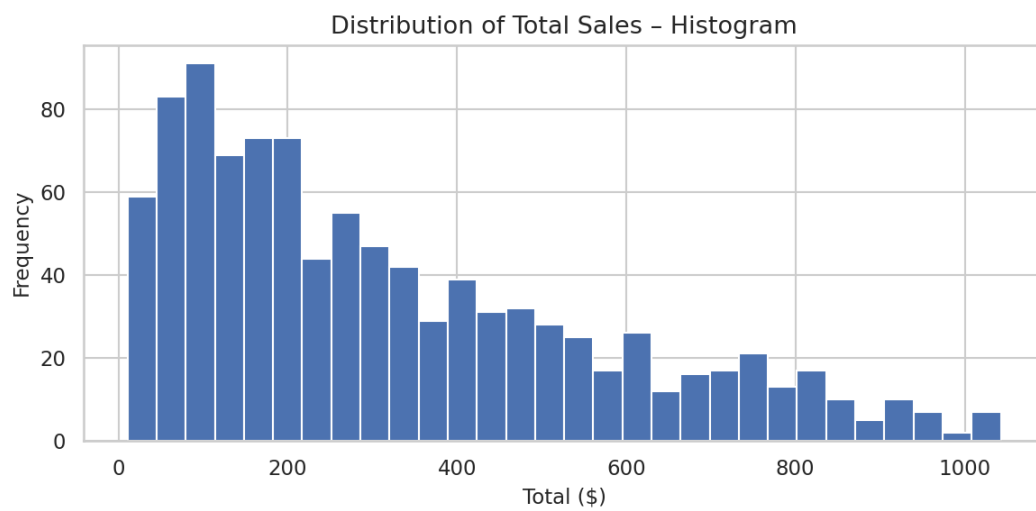


Figure: Histogram

5.7 KDE / Density Plot

The KDE plot smooths the Total distribution. It confirms the right skew and reveals a peak around \$100–\$200. The gradual right tail represents high-value transactions from premium products or bulk buying.

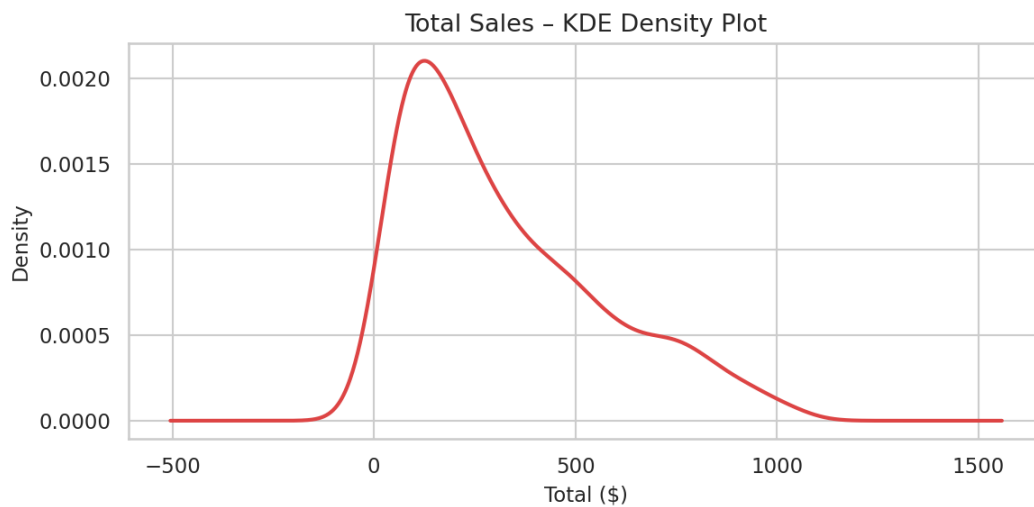


Figure: KDE / Density Plot

5.8 Box Plot (Pandas)

Side-by-side box plots for Total grouped by product line reveal that Electronic Accessories and Fashion Accessories have the widest spread (most variable spend). Home & Lifestyle has the highest median transaction value.

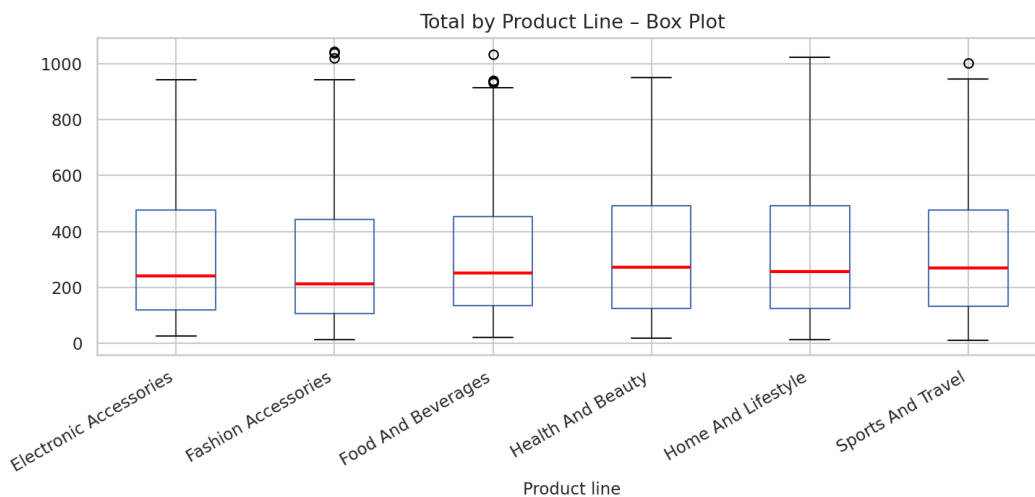


Figure: Box Plot (Pandas)

5.9 Hexbin Plot

The hexbin plot reinforces the Unit Price vs Total relationship. The densest regions (darkest hexagons) appear at low-to-mid price and low-to-mid total values, representing the most common transaction type.

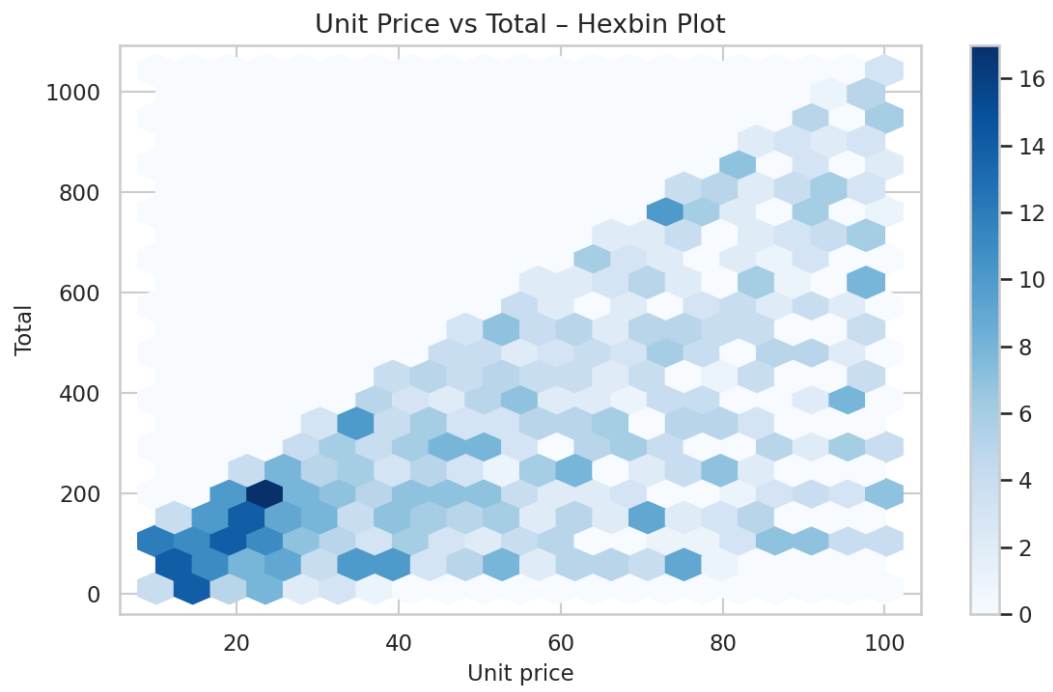


Figure: Hexbin Plot

5.10 Pie Chart

The payment method pie chart shows an almost perfectly three-way split: Ewallet (34.5%), Cash (34.4%), and Credit Card (31.1%). This suggests customers have no strong payment preference, which has operational implications for checkout infrastructure.

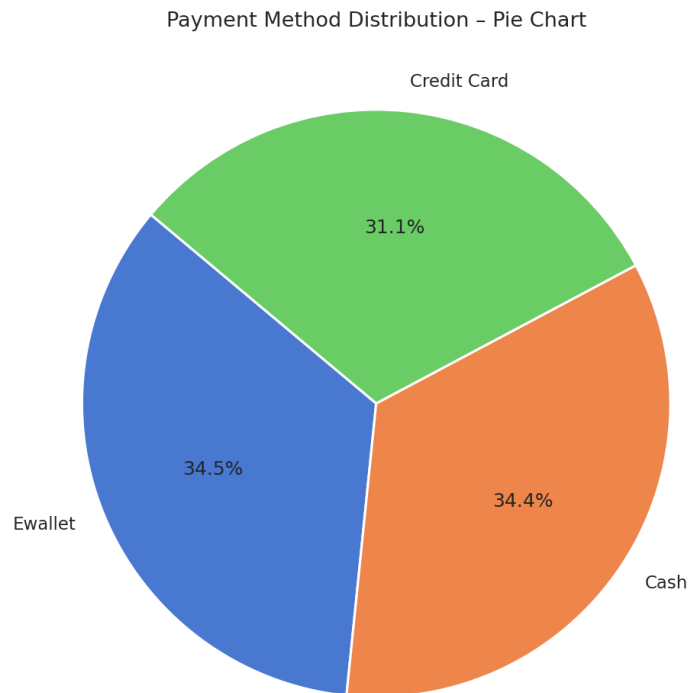


Figure: Pie Chart

6. Data Visualization – Seaborn Methods

14 visualizations using Seaborn's distribution, relationship, categorical & matrix APIs

6.1 Distribution Plots

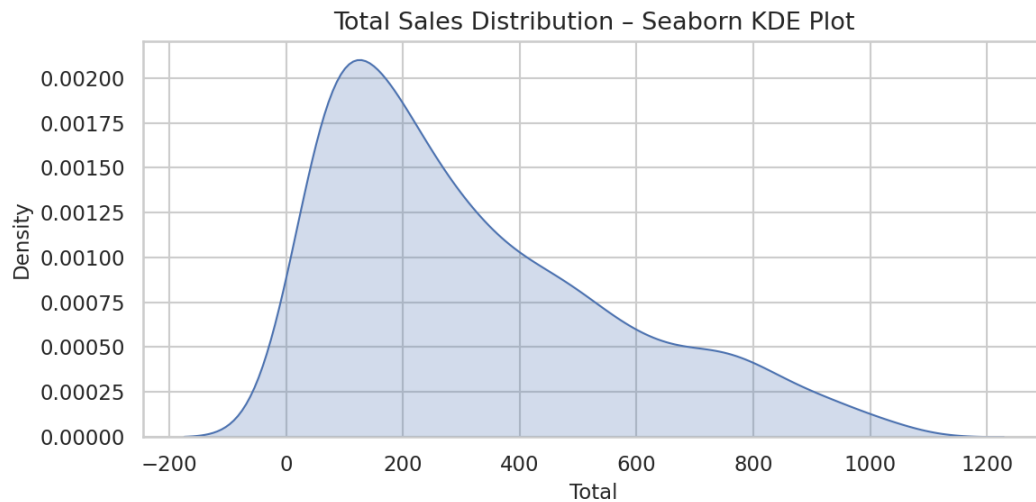


Figure: Seaborn KDE Plot – Total Sales Distribution

The Seaborn KDE plot adds a shaded fill beneath the density curve, clearly showing the right-skewed nature of Total. The mode (~\$150) is well below the mean (~\$323), confirming positive skewness.

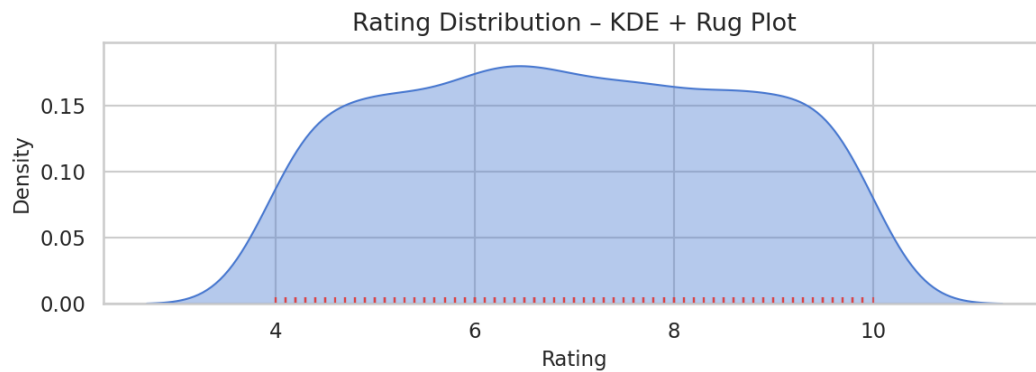


Figure: KDE + Rug Plot – Rating Distribution

The rug plot layered beneath the Rating KDE shows individual data points. The near-uniform distribution of ratings from 4.0 to 10.0 suggests customers rate the supermarket consistently across the full available scale.

6.2 Relationship Plots

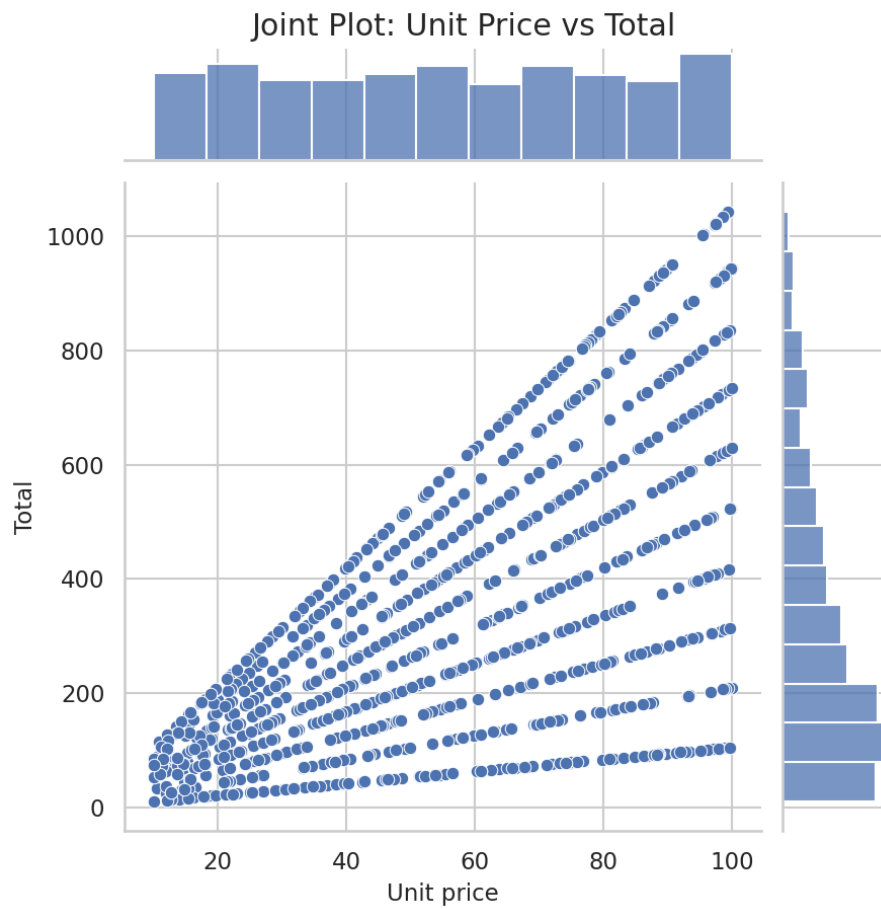


Figure: Joint Plot – Unit Price vs Total with marginal distributions

The joint plot combines scatter (center) with marginal histograms (edges). The strong diagonal pattern confirms the positive correlation between unit price and total spend. Marginal distributions show unit price is near-uniform while total is right-skewed.

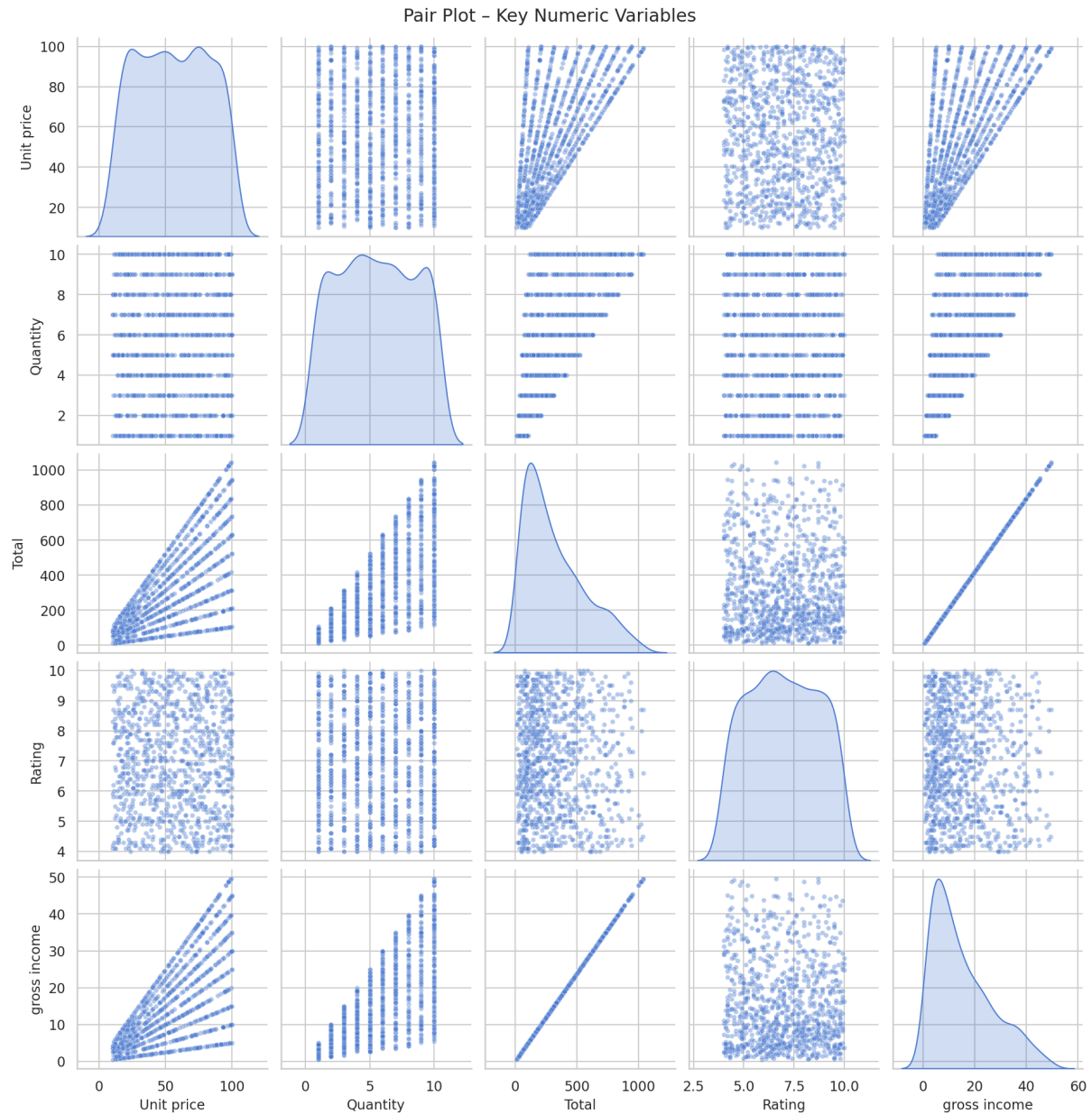


Figure: Pair Plot – All pairwise relationships among key numeric variables

The pair plot reveals the following key relationships at a glance: (1) Total–cogs have a near-perfect linear relationship (they are mathematically linked). (2) Unit price and Total show moderate positive correlation. (3) Rating is uncorrelated with all financial variables — customer satisfaction is independent of spend.

6.3 Categorical Plots

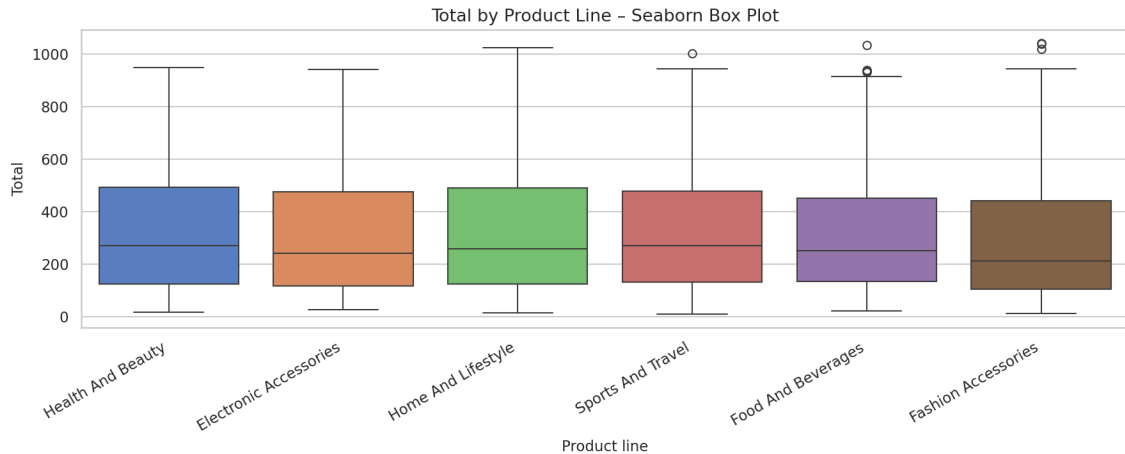


Figure: Seaborn Box Plot – Total by Product Line

This seaborn box plot offers richer formatting and better aesthetics than the Pandas equivalent. Outlier dots visible above each whisker represent the 9 high-value transactions identified in the IQR analysis. Electronic Accessories shows the highest single outlier (~\$1,042).

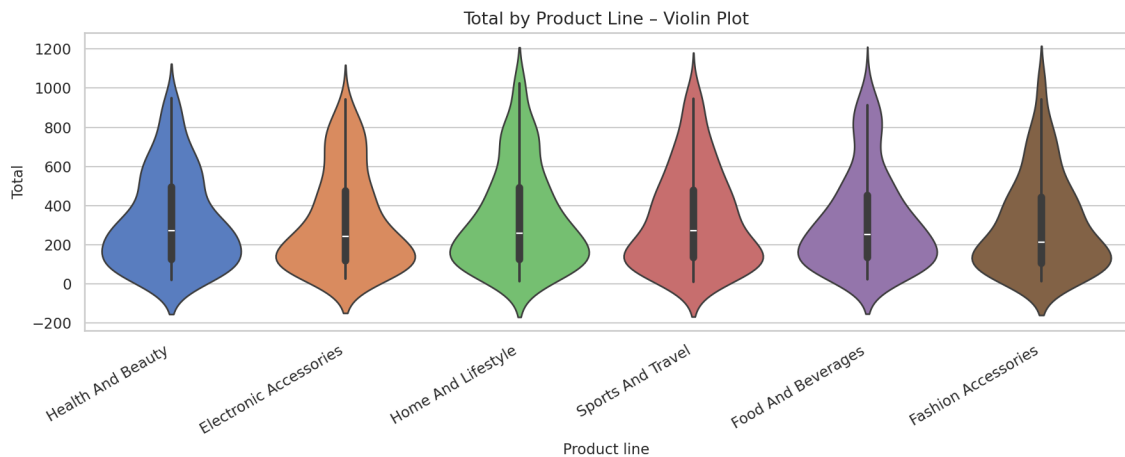


Figure: Violin Plot – Total by Product Line

Violin plots combine box plot statistics with KDE distribution shapes. All product lines show a bimodal-like pattern: a large mass of smaller transactions and a smaller bulge at higher values. Electronic Accessories has the widest violin, confirming its high variance in transaction size.

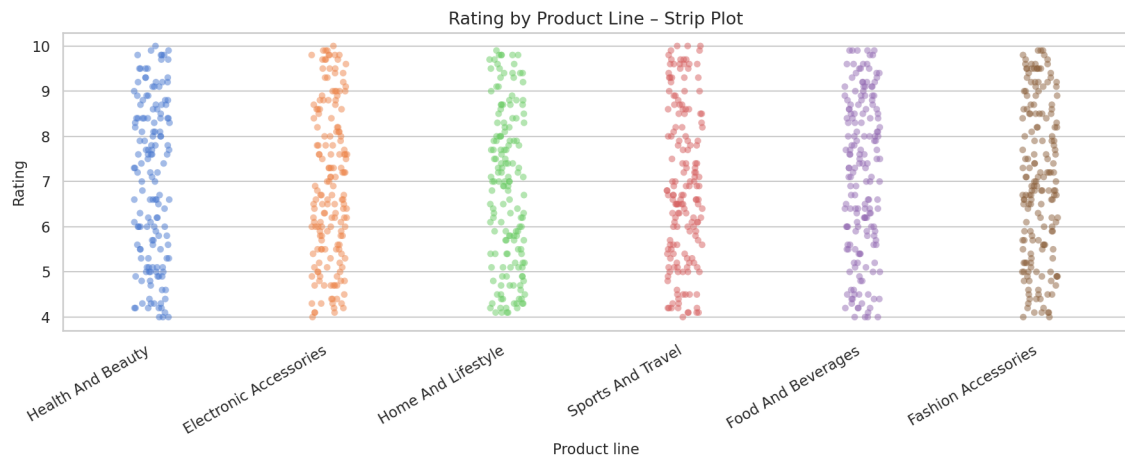


Figure: Strip Plot – Rating by Product Line

The strip plot shows individual customer ratings for each product line. All lines receive a spread of ratings from 4 to 10, with no product line systematically rated higher or lower. This suggests product type does not influence customer satisfaction rating.

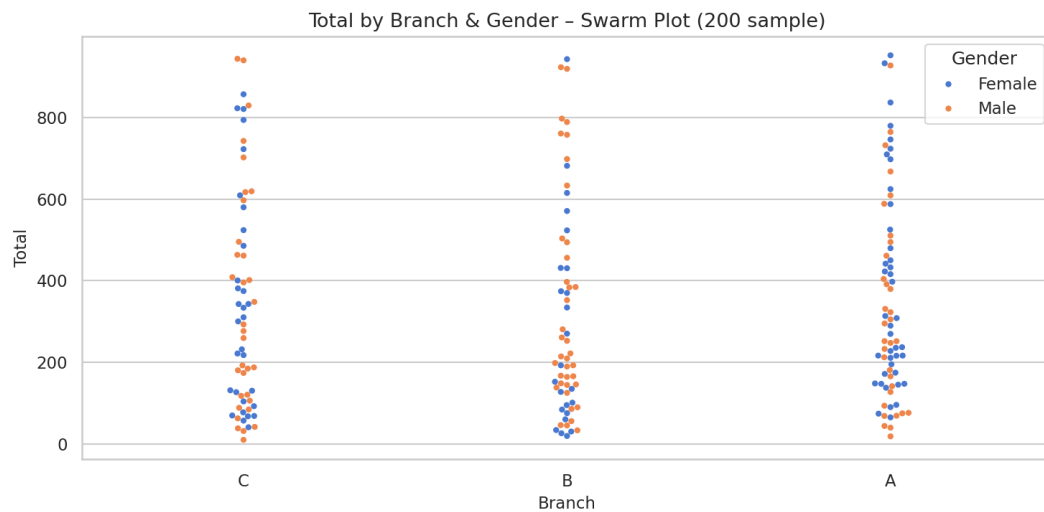


Figure: Swarm Plot – Total by Branch & Gender (200-row sample)

The swarm plot (200-row sample) shows no systematic difference in spending patterns between male and female customers across branches. The distributions overlap substantially in all three branches, confirming gender-neutral purchasing behaviour.

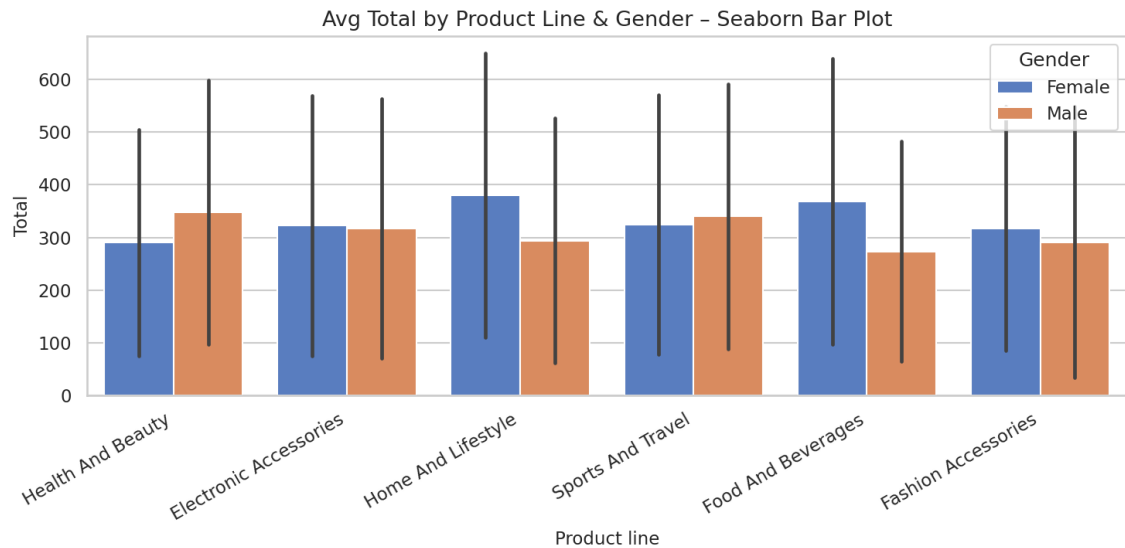


Figure: Seaborn Bar Plot – Avg Total by Product Line & Gender

This bar plot with error bars (± 1 SD) compares average transaction total by product line and gender. Female customers spend slightly more on Fashion Accessories and Food & Beverages; males spend slightly more on Health & Beauty and Sports & Travel. However, the differences are within one standard deviation and likely not statistically significant.

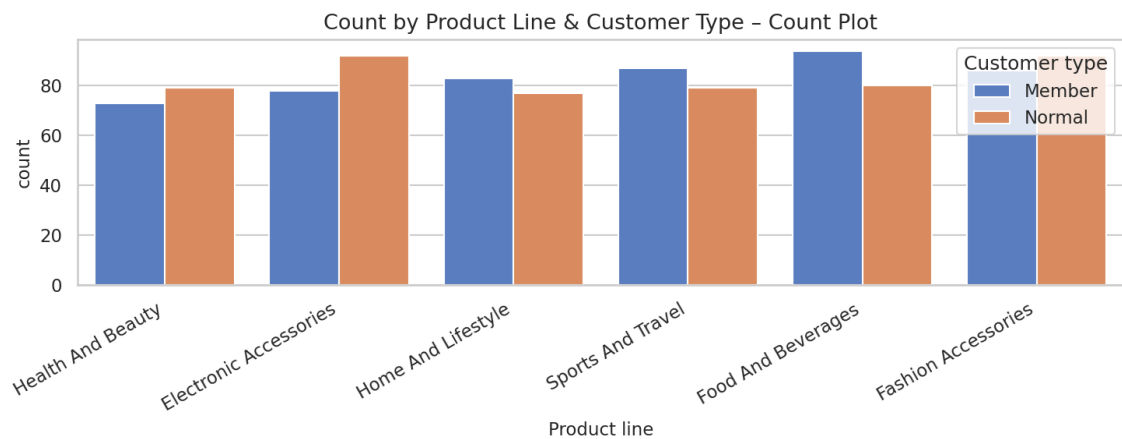


Figure: Count Plot – Transaction Count by Product Line & Customer Type

The count plot shows how transactions are distributed across product lines by customer type. Members and Normal customers purchase in roughly equal proportions across all categories. Fashion Accessories has the most transactions overall, while Health & Beauty has the fewest.

6.4 Matrix Plot

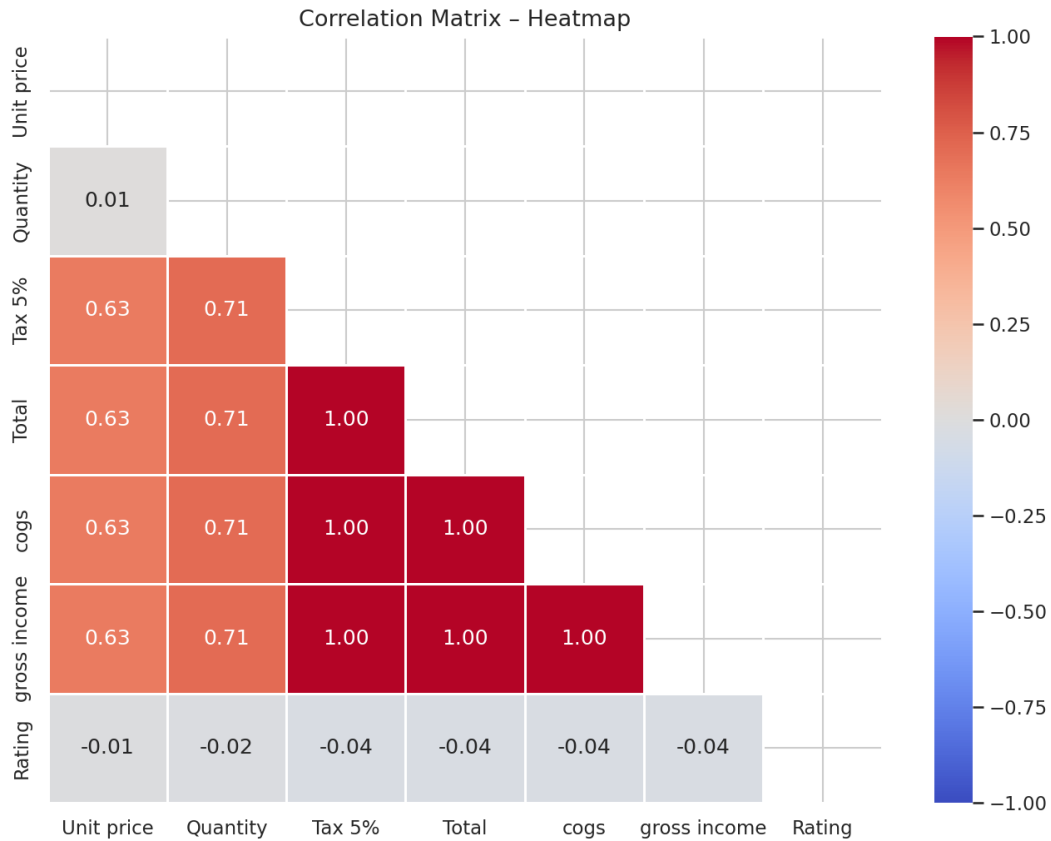


Figure: Correlation Heatmap – Numeric Variables

The correlation heatmap (lower triangle) reveals key relationships. Total, cogs, Tax 5%, and gross income are all perfectly correlated with each other ($r \approx 1.0$) because they are mathematically derived from the same base. Unit Price and Quantity both have moderate positive correlations with Total (~ 0.63 and ~ 0.71 respectively). Crucially, Rating has near-zero correlation with all financial metrics, confirming that customer satisfaction is independent of purchase size.

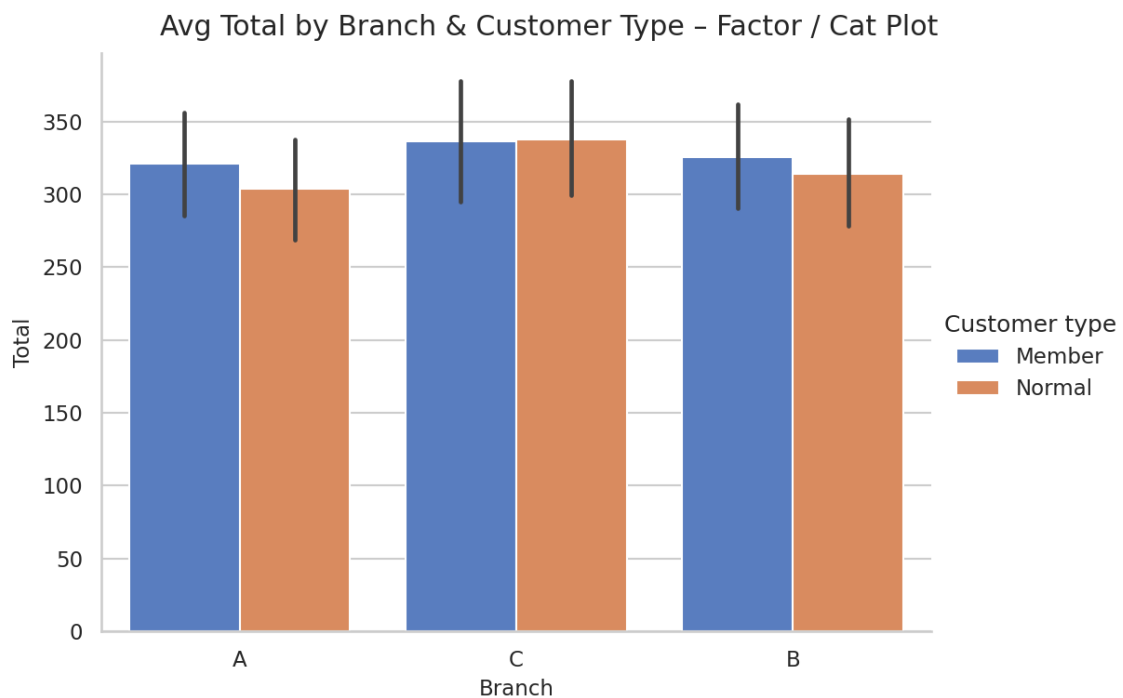


Figure: Factor / Cat Plot – Avg Total by Branch & Customer Type

The factor plot (catplot) shows that Branch C (Naypyitaw) has the highest average transaction total for both Member and Normal customers. Members tend to spend slightly more than Normal customers across all branches, which may reflect access to member discounts driving larger basket sizes.

7. Key Insights & Conclusions

7.1 Business Insights

Revenue Performance	Total revenue over 3 months was \$322,967 across 1,000 transactions, averaging \$323 per sale. Branch C (Naypyitaw) is the top-performing branch by total revenue despite similar transaction counts.
Top Product Line	Food & Beverages generated the highest cumulative revenue (\$56,145). However, Health & Beauty has the highest average transaction value, suggesting premium pricing in that category.
Payment Trends	Digital payments (Ewallet + Credit Card) account for ~66% of transactions, indicating strong digital payment adoption. Ewallet is the single most popular method at 34.5%.
Customer Segments	Member and Normal customers are nearly equally split (50/50). Members show slightly higher average spend, suggesting loyalty programs are working but have room to deepen impact.
Gender Balance	Male and female customers are nearly equally represented with no significant difference in spending patterns. Marketing campaigns can treat both segments similarly.
Customer Satisfaction	Average rating of 6.97/10 is decent but not exceptional. Ratings are uncorrelated with spend, suggesting service quality improvements are needed across all product lines equally.
Peak Hours	Data includes time-of-day information. Transactions occur between 10 AM and 9 PM, which can inform staffing decisions at peak shopping hours.

7.2 Data Quality Conclusions

The dataset was already in excellent quality. No missing values or duplicates were found. The only processing required was date/time type conversion and text normalization. The 9 IQR outliers in Total/cogs are genuine high-value transactions (not errors) and were retained to preserve analytical accuracy.

7.3 Statistical Conclusions

- The distributions of Unit Price and Quantity are near-uniform — the store stocks products across all price tiers.
- Total sales is right-skewed with positive outliers — a small number of high-value purchases significantly impact revenue.
- Total, cogs, Tax 5%, and gross income are mathematically identical in correlation — only one needs to be tracked for financial KPIs.

- Rating follows a near-normal distribution; the average (6.97) suggests customers are generally satisfied but not delighted.
- There is no seasonal trend within this 3-month window — a longer time series would be needed to detect seasonality.

7.4 Recommendations

Loyalty Programs	Deepen member benefits to widen the spending gap vs Normal customers — current difference is minimal.
Product Focus	Promote Health & Beauty more aggressively given its high average transaction value.
Rating Improvement	Launch post-purchase feedback initiatives targeting product lines with lower median ratings.
Branch C Replication	Study what makes Branch C (Naypyitaw) the highest revenue generator and replicate in Branches A & B.
Peak Hour Staffing	Use hourly transaction data to optimise cashier staffing during peak shopping windows.
Digital Payment Growth	Promote Credit Card and Ewallet payment with cashback incentives to reduce cash-handling costs.